

SOLUTION We first compute the gradient vector at $(2, -1)$:

$$\nabla f(x, y) = 2xy^3 \mathbf{i} + (3x^2y^2 - 4) \mathbf{j}$$

$$\nabla f(2, -1) = -4 \mathbf{i} + 8 \mathbf{j}$$

Note that $\mathbf{v}$ is not a unit vector, but since $|\mathbf{v}| = \sqrt{29}$, the unit vector in the direction of $\mathbf{v}$ is

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{2}{\sqrt{29}} \mathbf{i} + \frac{5}{\sqrt{29}} \mathbf{j}$$

Therefore, by Equation 9, we have

$$\begin{aligned} D_{\mathbf{u}}f(2, -1) &= \nabla f(2, -1) \cdot \mathbf{u} = (-4 \mathbf{i} + 8 \mathbf{j}) \cdot \left(\frac{2}{\sqrt{29}} \mathbf{i} + \frac{5}{\sqrt{29}} \mathbf{j} \right) \\ &= \frac{-4 \cdot 2 + 8 \cdot 5}{\sqrt{29}} = \frac{32}{\sqrt{29}} \end{aligned}$$



Problem: Find the shortest distance from the point $(2, 0, -3)$ to the plane $x + y + z = 1$.

Solution:

The distance d from a point on the plane (x, y, z) to the given point $(2, 0, -3)$ is:

$$d = \sqrt{(x-2)^2 + y^2 + (z+3)^2}$$

Since $x + y + z = 1$, we can express z in terms of x and y :

$$z = 1 - x - y$$

Substitute this into the distance formula to get a function in terms of x and y :

$$d = \sqrt{(x-2)^2 + y^2 + (1-x-y+3)^2}$$

To simplify the calculations, we minimize d^2 instead of d . Let $f(x, y) = d^2$:

$$f(x, y) = (x-2)^2 + y^2 + (4-x-y)^2$$

Expand $f(x, y)$:

$$\begin{aligned} f(x, y) &= (x-2)^2 + y^2 + (4-x-y)^2 \\ &= (x^2 - 4x + 4) + y^2 + (16 - 8x - 8y + x^2 + 2xy + y^2) \end{aligned}$$

Combine like terms:

$$f(x, y) = 2x^2 + 2y^2 + 2xy - 12x - 8y + 20$$

To find the minimum distance, we need to find the critical points by setting the partial derivatives of $f(x, y)$ with respect to x and y to zero.

Partial derivatives:

$$f_x = 4x + 2y - 12$$

$$f_y = 2x + 4y - 8$$

Set $f_x = 0$ and $f_y = 0$:

$$4x + 2y = 12$$

$$2x + 4y = 8$$

Solving these equations: From the first equation:

$$2x + y = 6$$

From the second equation:

$$x + 2y = 4$$

Solving this system, we find:

$$x = \frac{8}{3}, \quad y = \frac{2}{3}$$

1

Calculate z at the Critical Point:

$$z = 1 - x - y = 1 - \frac{8}{3} - \frac{2}{3} = -\frac{7}{3}$$

So, the point on the plane closest to $(2, 0, -3)$ is $(\frac{8}{3}, \frac{2}{3}, -\frac{7}{3})$.

Calculate the Distance:

$$\begin{aligned} d &= \sqrt{\left(2 - \frac{8}{3}\right)^2 + \left(0 - \frac{2}{3}\right)^2 + \left(-3 + \frac{7}{3}\right)^2} \\ &= \sqrt{\left(-\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^2} \\ &= \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{4}{9}} = \sqrt{\frac{12}{9}} = \sqrt{\frac{4}{3}} = \frac{2\sqrt{3}}{3} \end{aligned}$$

Final Answer: The shortest distance from the point $(2, 0, -3)$ to the plane $x + y + z = 1$ is $\frac{2\sqrt{3}}{3}$.

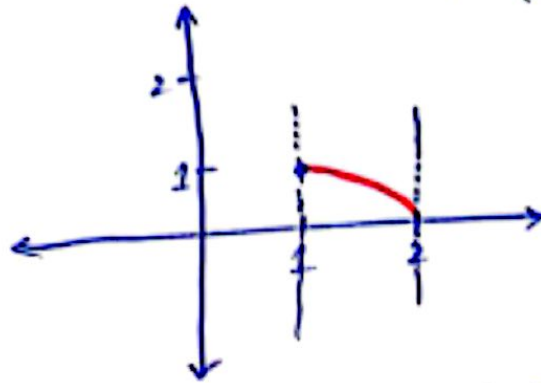
Q1 :

Solution :

$$y = \sqrt{2x - x^2}$$

$$\Rightarrow y^2 = 2x - x^2$$

$$\Rightarrow (x-1)^2 + y^2 = 1$$



The right-half of ~~semi-circle~~ *quarter circle* of radius 1 centered at (1, 0)

$$\int_1^2 \int_0^{\sqrt{2x-x^2}} \frac{1}{(x^2+y^2)^2} dy dx = \int_0^{\frac{\pi}{4}} \int_{\frac{1}{\cos \theta}}^{2 \cos \theta} \frac{r}{r^4} dr d\theta \longrightarrow 5$$

$$= \int_0^{\frac{\pi}{4}} \int_{\frac{1}{\cos \theta}}^{2 \cos \theta} \frac{1}{r^3} dr d\theta$$

$$= \int_0^{\frac{\pi}{4}} \left[\frac{-1}{2r^2} \right]_{\frac{1}{\cos \theta}}^{2 \cos \theta} d\theta$$

$$= \frac{\pi}{16} \longrightarrow 5$$

SOLUTION Here the change of variables is given by

$$x = \rho \sin \phi \cos \theta \quad y = \rho \sin \phi \sin \theta \quad z = \rho \cos \phi$$

We compute the Jacobian as follows:

$$\begin{aligned} \frac{\partial(x, y, z)}{\partial(\rho, \theta, \phi)} &= \begin{vmatrix} \sin \phi \cos \theta & -\rho \sin \phi \sin \theta & \rho \cos \phi \cos \theta \\ \sin \phi \sin \theta & \rho \sin \phi \cos \theta & \rho \cos \phi \sin \theta \\ \cos \phi & 0 & -\rho \sin \phi \end{vmatrix} \\ &= \cos \phi \begin{vmatrix} -\rho \sin \phi \sin \theta & \rho \cos \phi \cos \theta \\ \rho \sin \phi \cos \theta & \rho \cos \phi \sin \theta \end{vmatrix} - \rho \sin \phi \begin{vmatrix} \sin \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \sin \phi \cos \theta \end{vmatrix} \\ &= \cos \phi (-\rho^2 \sin \phi \cos \phi \sin^2 \theta - \rho^2 \sin \phi \cos \phi \cos^2 \theta) \\ &\quad - \rho \sin \phi (\rho \sin^2 \phi \cos^2 \theta + \rho \sin^2 \phi \sin^2 \theta) \\ &= -\rho^2 \sin \phi \cos^2 \phi - \rho^2 \sin \phi \sin^2 \phi = -\rho^2 \sin \phi \end{aligned}$$

Since $0 \leq \phi \leq \pi$, we have $\sin \phi \geq 0$. Therefore

$$\left| \frac{\partial(x, y, z)}{\partial(\rho, \theta, \phi)} \right| = |-\rho^2 \sin \phi| = \rho^2 \sin \phi$$

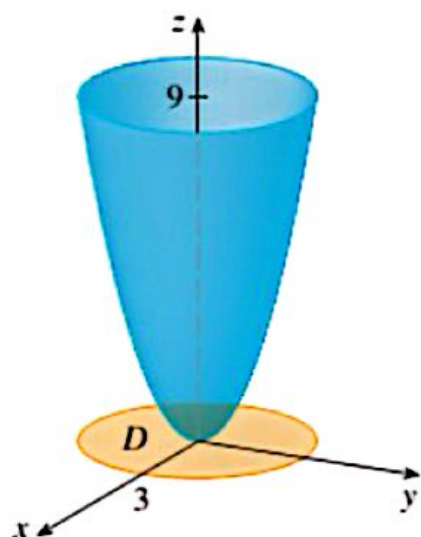


FIGURE 5

SOLUTION The plane intersects the paraboloid in the circle $x^2 + y^2 = 9$, $z = 9$. Therefore the given surface lies above the disk D with center the origin and radius 3. (See Figure 5.) Using Formula 3, we have

$$\begin{aligned} A &= \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA = \iint_D \sqrt{1 + (2x)^2 + (2y)^2} dA \\ &= \iint_D \sqrt{1 + 4(x^2 + y^2)} dA \end{aligned}$$

Converting to polar coordinates, we obtain

$$\begin{aligned} A &= \int_0^{2\pi} \int_0^3 \sqrt{1 + 4r^2} r dr d\theta = \int_0^{2\pi} d\theta \int_0^3 \frac{1}{8} \sqrt{1 + 4r^2} (8r) dr \\ &= 2\pi \left(\frac{1}{8} \right) \frac{2}{3} (1 + 4r^2)^{3/2} \Big|_0^3 = \frac{\pi}{6} (37\sqrt{37} - 1) \end{aligned}$$